

Name: _____

Reg. No.: _____

Date: 5 / 6 / 2024

Period: 2 hours



REPUBLIC OF YEMEN

Aden University

Faculty of Engineering

Chemistry
Final Exam
For B1C/B1M/B1MAR -E
2023-2024

Write all the questions below

Q.1 Choose the correct answers (10 marks):

Condensation and evaporation are _____.

(1) types of precipitation, (2) phase changes, (3) types of chemical reactions, (4) absorption processes

Which is an extensive property of a substance?

(1) Color (2) Boiling point (3) Volume (4) Density

A physical change to convert gas to liquid is _____.

(1) sublimation, (2) freezing, (3) evaporation, (4) condensation

Chemical reactions have several types such as _____.

(1) decomposition, (2) redox reactions, (3) 1 & 2, (4) all are wrong

If $\Delta H(+)$, $\Delta S(+)$ then ΔG _____.

(1) (+) at high T, (2) (+) at low T, (3) spontaneous at high T, (4) spontaneous at all T

Q.2 Complete the table below (10 marks):

Cation	Anion	Chemical formula	Systematic name
Al^{+3}	SO_3^{2-}		
NH_4^{+1}	CO_3^{2-}		
Fe^{+3}	ClO_4^-		
H^+	IO^-		
Na^+	S^{2-}		

Q.3 Give the order of the following (10 marks):

First ionization energy: ${}_3Li$, ${}_7N$, ${}_9F$, ${}_{11}Na$

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Electronegativity: ${}_{20}Ca$, ${}_{37}Rb$, ${}_9F$, ${}_{15}P$

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Electron Configurations: $_{27}\text{Co}$, $_{33}\text{As}$, $_{18}\text{Ar}$, $_{14}\text{Si}$

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The entropy (S) of the gases: $_{54}\text{Xe}$, $_{18}\text{Ar}$, $_{2}\text{He}$, $_{36}\text{Kr}$

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The frequency: $2.5 \times 10^4 \text{ MHz}$, 25 GHz , $25 \times 10^6 \text{ Hz}$, 250 KHz

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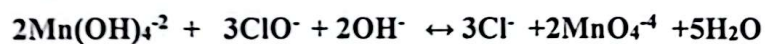
Q.4 CHOOSE ONLY TWO

Perform the following calculations (10 marks each)

(1) Complete and balance the following reactions:



(2) (a) How many atoms, moles and mass of each element in the reaction



(A.wt: $\text{Mn}=55$, $\text{O}=16$, $\text{H}=1$, $\text{Cl}=35.5$)

(b) Carry out the following mathematical operations and give the answers in the correct number of significant figures: $(1.33 \div 2.8) - 8.41 = ?$

(3) Calculate the mass percentage of oxygen in 4g CaCO_3 if CaCO_3 has a molar mass of 100 g/mol and oxygen's mass is 16 g/mol.

*ملحوظة: يسمح باستخدام الآلة الحاسبة وليس الجوال Only a normal calculator is allowed

Good Luck
Dr. Adel A.M. Saeed